Venkata Lakshmi Parimala Pasupuleti

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Virginia, USA (Remote)

PROFESSIONAL SUMMARY

Applied AI Engineer with 8+ years of production engineering in real estate and fintech technology and hands-on experience building, deploying, and evaluating agentic AI systems from scratch. I've built LeaseI — an 8-agent LangGraph pipeline with RAG, embedding pipelines, and human-in-the-loop escalation for housing compliance operations (93.6% precision on 1,240 test cases) — and AgentProbe, an open-source LLM evaluation framework with a 5-metric eval engine, CI-integrated regression gate, and multi-provider LLM router. At Entrata (Housing Technology SaaS), I embedded cross-functionally with operations, product, and business teams to translate workflow bottlenecks into production AI solutions — the same motion as AI discovery & strategy embedded in a product org. Self-directed by default: I defined both projects from problem identification through architecture, build, and production evaluation — no PM, no roadmap. Strong in Python, TypeScript, RAG pipelines, agentic frameworks, vector databases, and evaluation-driven AI delivery. AgentProbe: github.com/venkatalakshmiparimala/AgentProbe

CORE COMPETENCIES

AI Discovery & Strategy | Agentic AI & LLM-Powered Products | RAG Pipelines & Embedding Systems | LLM Evaluation Frameworks | Cross-Functional AI Embedding | Production AI Deployment & Monitoring | Real Estate & PropTech Domain | Self-Directed AI Ownership

WORK EXPERIENCE

Blue Horse Digital

Sep 2026 – Present

AI/ML Engineer (Volunteer) — Virginia, USA

- Building agentic AI pipelines and ML model workflows; integrating data systems with product-facing APIs; translating raw business workflows into AI-powered features with measurable outcomes.

Magizhs Tech Solution

May 2025 – Aug 2025

AI/ML Engineer Intern (Forward Deployed)

- **AI discovery & delivery embedded in operations:** led discovery sessions to identify where AI created real workflow leverage, built a LangGraph agentic pipeline (parse → plan → retrieve → validate → routing), integrated with client enterprise systems via REST APIs, deployed guardrails enforcing business rules before output delivery; owned the technical narrative from use case identification through production rollout.
- Implemented RAG pipeline with cited compliance decisions; translated implementation learnings into reusable product components; served as the technical bridge between AI capabilities and client proof-of-value.

George Mason University

Aug 2024 – May 2026

Graduate Teaching Assistant + Research Associate

- Built multi-source prediction pipelines (Python, R, $R^2 = 0.977$); fine-tuned a custom classification model (HuggingFace + PyTorch); applied Isolation Forest for automated data integrity checks across 3 research pipelines; designed Tableau and ggplot2 dashboards for faculty leadership (40% comprehension improvement).

Entrata (Housing Technology SaaS — 100K+ monthly active users)

Jan 2017 – Aug 2024

Senior Software Engineer — Lehi, UT (Remote)

- **8 years owning production systems in real estate technology:** built and maintained the full integration layer connecting property management, leasing, eligibility, and compliance workflows across 5+ enterprise systems — REST APIs, webhook endpoints, event-driven async pipelines (RabbitMQ), CRM connectors, and third-party data integrations for 100K+ monthly active users at 99.5% uptime. Direct domain overlap with Splitero's real estate + financial data environment.
- **Cross-functional AI discovery and automation:** embedded with operations and product teams to identify where recurring manual workflows created leverage opportunities; translated patterns into automated pipeline features; built Python/PL-SQL automation eliminating 80% of manual operational workload; maintained a bug-to-backlog loop feeding structured product feedback from field learnings — the same "identify, scope, deliver, iterate" motion this role requires.
- Optimized 200+ SQL queries (30% performance gain); implemented AWS S3 document migration (100% PHI compliance, zero security incidents); designed schemas for 20+ production projects in multi-tenant PostgreSQL environments; 300+ peer code reviews; elevated team engineering standards across 120+ Agile sprints.

PROJECTS

Leasel: Applied AI Platform for Real Estate & Housing Operations

LangGraph — RAG — FastAPI — LLM Evaluation — HITL

Designed and built an end-to-end applied AI system for housing compliance operations: 8-agent LangGraph pipeline with durable state, conditional routing, and human-in-the-loop escalation (Triage → Retrieval → Configuration → Historical → Structured Data → Reasoning → Validation → Escalation). RAG pipeline with hybrid TF-IDF + semantic retrieval, embedding pipelines, citation grounding, and document intelligence. Multi-system orchestration across structured databases, document stores, configuration systems, and regulatory rule engines. Results: 93.6% precision / 87.9% F1 on 1,240 test cases; 100% citation correctness. Scoped, built, and evaluated entirely solo — no PM, no roadmap.

github.com/venkatalakshmiparimala/Lease_complaince_rag

AgentProbe: Open-Source LLM Evaluation Framework

FastAPI — LangGraph — ChromaDB — Multi-Provider LLM — CI/CD

Production evaluation toolkit for AI pipelines: 15-endpoint FastAPI backend, 5-metric eval engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression gate with golden datasets, multi-provider LLM router (OpenAI/Anthropic/Groq/Ollama), vector database (ChromaDB), and observability dashboard. Built to close the agent performance feedback loop — used to evaluate Leasel across 1,240+ test cases before production. Zero vendor lock-in, fully local.

github.com/venkatalakshmiparimala/AgentProbe

AI on FHIR: Production Conversational AI Integration for Clinical Systems

Flask — LLMs — BERT — REST API — HIPAA

Conversational AI platform translating natural-language queries into FHIR API calls using LLMs, BERT, and embeddings — direct API integration with clinical systems, HIPAA-compliant RBAC, NLP pipeline for multi-system data discovery across EN/ES. ~60% reduction in manual retrieval time. Production deployment with SOC 2-aligned data handling. Full AI development lifecycle ownership: prototype → evaluation → deployment → monitoring.

github.com/venkatalakshmiparimala/AI_on_FHIR

EDUCATION

M.S. Data Analytics Engineering — George Mason University — May 2026

GPA 3.87 — Virginia, USA

B.Tech Computer Science & Engineering — JNTU Anantapur — 2017

SKILLS

AI & Agentic Systems: LangChain, LangGraph, Agentic AI, Multi-Agent Orchestration, LLM APIs (OpenAI, Anthropic, Groq), RAG Pipelines, Embedding Pipelines, Vector Databases (ChromaDB), Prompt Engineering, Function Calling, HITL, Evaluation Frameworks, LLM Evaluation, Hallucination Detection, Groundedness, AI Observability, Guardrails, MCP, FastAPI, Workflow Automation

Languages & Frameworks: Python, TypeScript, JavaScript, Node.js, React.js, SQL, FastAPI, Flask, Django, Java, Spring Boot, Bash

Cloud & Infrastructure: AWS (S3, EC2, Lambda, Glue), PostgreSQL, Redis, Docker, Kubernetes, CI/CD (GitHub Actions), Linux

Data & Integration: REST APIs, Webhook Endpoints, RabbitMQ, Event-Driven Architecture, ETL Pipelines, OAuth, OpenAPI, CRM Connectors, Data Modeling, Feature Engineering, scikit-learn, PyTorch